

Started on Thursday, 13 August 2026, 10:55 AM

State Finished

Completed on Thursday, 13 August 2026, 11:25 AM

Time taken 29 mins 58 secs

Grade 10.00 out of 10.00 (100%)

Insert all books from `Out_of_print_books` into `Books`

Table attributes are `ISBN`, `Title`, `Author`, `Publisher`, `YearPublished`

For example:

Test	Result				
select * from Books;	ISBN	Title	Author	Publisher	
	YearPublished				
	-----	-----	-----	-----	-----

	978-1234567890	The Lost World	Arthur Conan Doyle	Vintage Books	1912
	978-0987654321	Gone with the	Margaret Mitchell	Macmillan	1936
	978-1122334455	Moby Dick	Herman Melville	Harper & Brot	1851

Answer: (penalty regime: 0 %)

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```
INSERT INTO Books (ISBN, Title, Author, Publisher, YearPublished)
SELECT ISBN, Title, Author, Publisher, YearPublished
FROM Out_of_print_books;
```

	Test	Expected	Got	
✓	select * from Books;	ISBN Title Author Publisher YearPublished ----- ----- ----- 978-1234567890 The Lost World Arthur Conan Doyle Vintage Books 1912 978-0987654321 Gone with the Margaret Mitchell Macmillan 1936 978-1122334455 Moby Dick Herman Melville Harper & Brot 1851	ISBN Title Author Publisher YearPublished ----- ----- ----- 978-1234567890 The Lost World Arthur Conan Doyle Vintage Books 1912 978-0987654321 Gone with the Margaret Mitchell Macmillan 1936 978-1122334455 Moby Dick Herman Melville Harper & Brot 1851	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Create a new table named **contacts** with the following specifications:

- **contact_id** as INTEGER and primary key.
- **first_name** as TEXT and not NULL.
- **last_name** as TEXT and not NULL.
- **email** as TEXT.
- **phone** as TEXT and not NULL with a check constraint to ensure the length of **phone** is at least 10 characters.

For example:

Test	Result
<pre>INSERT INTO contacts (contact_id, first_name, last_name, email, phone) VALUES (1, 'John', 'Doe', 'john.doe@example.com', '1234567890'); SELECT * FROM contacts;</pre>	<pre>contact_id first_name last_name email phone ----- 1 John Doe john.doe@example.com 1234567890</pre>

Answer: (penalty regime: 0 %)

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```
CREATE TABLE contacts (
    contact_id INTEGER PRIMARY KEY,
    first_name TEXT NOT NULL,
    last_name TEXT NOT NULL,
    email TEXT,
    phone TEXT NOT NULL,
    CHECK (LENGTH(phone) >= 10)
);
```

	Test	Expected	Got	
✓	<pre>INSERT INTO contacts (contact_id, first_name, last_name, email, phone) VALUES (1, 'John', 'Doe', 'john.doe@example.com', '1234567890'); SELECT * FROM contacts;</pre>	<pre>contact_id first_name last_name email phone ----- 1 John Doe john.doe@example.com 1234567890</pre>	<pre>contact_id first_name last_name email phone ----- 1 John Doe john.doe@example.com 1234567890</pre>	✓
✓	<pre>INSERT INTO contacts (contact_id, first_name, last_name, phone) VALUES (2, 'Jane', 'Smith', '0987654321'); select * from contacts;</pre>	<pre>contact_id first_name last_name email phone ----- 2 Jane Smith 0987654321</pre>	<pre>contact_id first_name last_name email phone ----- 2 Jane Smith 0987654321</pre>	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question 3

Correct

Mark 1.00 out of 1.00

Write an SQL query to add a new column **email** of type **TEXT** to the **Student_details** table, and ensure that this column cannot contain **NULL** values and make default value as 'Invalid'

For example:

Test	Result															
INSERT INTO Student_details (RollNo, Name, Gender, Subject, email) VALUES (1, 'John Doe', 'M', 'Math', 'john@example.com'); select * from Student_details;	<table><tr><td>RollN</td><td>Name</td><td>Gen</td><td>Subject</td><td>email</td></tr><tr><td>----</td><td>-----</td><td>---</td><td>-----</td><td>-----</td></tr><tr><td>1</td><td>John</td><td>M</td><td>Math</td><td>john@example.com</td></tr></table>	RollN	Name	Gen	Subject	email	----	-----	---	-----	-----	1	John	M	Math	john@example.com
RollN	Name	Gen	Subject	email												
----	-----	---	-----	-----												
1	John	M	Math	john@example.com												

Answer: (penalty regime: 0 %)

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```
ALTER TABLE Student_details  
ADD COLUMN email TEXT NOT NULL DEFAULT 'Invalid';
```

	Test	Expected	Got																															
✓	INSERT INTO Student_details (RollNo, Name, Gender, Subject, email) VALUES (1, 'John Doe', 'M', 'Math', 'john@example.com'); select * from Student_details;	<table><tr><td>RollN</td><td>Name</td><td>Gen</td><td>Subject</td><td>email</td></tr><tr><td>----</td><td>-----</td><td>---</td><td>-----</td><td>-----</td></tr><tr><td>1</td><td>John</td><td>M</td><td>Math</td><td>john@example.com</td></tr></table>	RollN	Name	Gen	Subject	email	----	-----	---	-----	-----	1	John	M	Math	john@example.com	<table><tr><td>RollN</td><td>Name</td><td>Gen</td><td>Subject</td><td>email</td></tr><tr><td>----</td><td>-----</td><td>---</td><td>-----</td><td>-----</td></tr><tr><td>1</td><td>John</td><td>M</td><td>Math</td><td>john@example.com</td></tr></table>	RollN	Name	Gen	Subject	email	----	-----	---	-----	-----	1	John	M	Math	john@example.com	✓
RollN	Name	Gen	Subject	email																														
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1	John	M	Math	john@example.com																														
RollN	Name	Gen	Subject	email																														
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1	John	M	Math	john@example.com																														

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Create a table named **Departments** with the following columns:

- **DepartmentID** as **INTEGER**
- **DepartmentName** as **TEXT**

For example:

Test	Result					
pragma table_info('Departments');	cid	name	type	notnull	dflt_value	pk
	----	-----	-----	-----	-----	---

	0	DepartmentID	INTEGER	0		0
	1	DepartmentName	TEXT	0		0

Answer: (penalty regime: 0 %)

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```
CREATE TABLE Departments (
    DepartmentID INTEGER,
    DepartmentName TEXT
);
```

	Test	Expected	Got																																																	
✓	pragma table_info('Departments');	<table><tr><td>cid</td><td>name</td></tr><tr><td>type</td><td>notnull</td></tr><tr><td>dflt_value</td><td>pk</td></tr><tr><td>-----</td><td>--</td></tr><tr><td>-----</td><td>----</td></tr><tr><td>-----</td><td>----</td></tr><tr><td>0</td><td>DepartmentID</td></tr><tr><td>INTEGER</td><td>0</td></tr><tr><td>0</td><td></td></tr><tr><td>1</td><td>DepartmentName</td></tr><tr><td>TEXT</td><td>0</td></tr><tr><td>0</td><td></td></tr></table>	cid	name	type	notnull	dflt_value	pk	-----	--	-----	----	-----	----	0	DepartmentID	INTEGER	0	0		1	DepartmentName	TEXT	0	0		<table><tr><td>cid</td><td>name</td></tr><tr><td>type</td><td>notnull</td></tr><tr><td>dflt_value</td><td>pk</td></tr><tr><td>-----</td><td>--</td></tr><tr><td>-----</td><td>----</td></tr><tr><td>-----</td><td>----</td></tr><tr><td>0</td><td>DepartmentID</td></tr><tr><td>INTEGER</td><td>0</td></tr><tr><td>0</td><td></td></tr><tr><td>1</td><td>DepartmentName</td></tr><tr><td>TEXT</td><td>0</td></tr><tr><td>0</td><td></td></tr></table>	cid	name	type	notnull	dflt_value	pk	-----	--	-----	----	-----	----	0	DepartmentID	INTEGER	0	0		1	DepartmentName	TEXT	0	0		✓
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1	DepartmentName																																																			
TEXT	0																																																			
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✓	INSERT INTO Departments (DepartmentID) VALUES (2); select * from Departments;	<table><tr><td>Depar</td><td>DepartmentName</td></tr><tr><td>-----</td><td>-----</td></tr><tr><td>2</td><td></td></tr></table>	Depar	DepartmentName	-----	-----	2		<table><tr><td>Depar</td><td>DepartmentName</td></tr><tr><td>-----</td><td>-----</td></tr><tr><td>2</td><td></td></tr></table>	Depar	DepartmentName	-----	-----	2		✓																																				
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Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Write an SQL query to add two new columns, `department_id` and `manager_id`, to the table `employee` with datatype of `INTEGER`. The `manager_id` column should have a default value of `NULL`.

For example:

Test	Result					
PRAGMA table_info(employee);	cid	name	type	notnull	dflt_value	pk
	-----	-----	-----	-----	-----	-----

	0	id	integer	0		0
	1	salary	number	0		0
	2	department	INTEGER	0		0
	3	manager_id	INTEGER	0	NULL	0

Answer: (penalty regime: 0 %)

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```
ALTER TABLE employee
ADD COLUMN department_id INTEGER;

ALTER TABLE employee
ADD COLUMN manager_id INTEGER DEFAULT NULL;
```

	Test	Expected			Got			
✓	PRAGMA table_info(employee);	cid	name	type	cid	name	type	✓
		notnull	dflt_value	pk	notnull	dflt_value	pk	
		-----	-----	---	-----	-----	---	
		---			---			
		0	id	integer	0	id	integer	
		0	0	0	0	0	0	
		1	salary	number	1	salary	number	
		0	0	0	0	0	0	
		2	department	INTEGER	2	department	INTEGER	
		0	0	0	0	0	0	
		3	manager_id	INTEGER	3	manager_id	INTEGER	
		0	0	NULL	0	0	NULL	
		0			0			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question 6

Correct

Mark 1.00 out of 1.00

Insert a new product with ProductID 101, Name Laptop, Category Electronics, Price 1500, and Stock 50 into the Products table.

For example:

Test	Result				
SELECT * FROM Products WHERE ProductID = 101;	ProductID	Name	Category	Price	
	Stock				
	-----	-----	-----	-----	----

	101	Laptop	Electronics	1500	50

Answer: (penalty regime: 0 %)

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```
INSERT INTO Products (ProductID, Name, Category, Price, Stock)
VALUES (101, 'Laptop', 'Electronics', 1500, 50);
```

	Test	Expected	Got																																																													
✓	SELECT * FROM Products WHERE ProductID = 101;	<table><tr><th>ProductID</th><th>Name</th><th>Category</th><th>Price</th><th>Stock</th></tr><tr><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>-----</td><td></td><td></td><td></td><td></td></tr><tr><td>-</td><td></td><td></td><td></td><td></td></tr><tr><td>101</td><td>Laptop</td><td></td><td></td><td></td></tr><tr><td>Electronics</td><td>1500</td><td>50</td><td></td><td></td></tr></table>	ProductID	Name	Category	Price	Stock	-----	-----	-----	-----	-----	-----					-					101	Laptop				Electronics	1500	50			<table><tr><th>ProductID</th><th>Name</th><th>Category</th><th>Price</th><th>Stock</th></tr><tr><td>-----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>-----</td><td></td><td></td><td></td><td></td></tr><tr><td>-</td><td></td><td></td><td></td><td></td></tr><tr><td>101</td><td>Laptop</td><td></td><td></td><td></td></tr><tr><td>Electronics</td><td>1500</td><td>50</td><td></td><td></td></tr></table>	ProductID	Name	Category	Price	Stock	-----	-----	-----	-----	-----	-----					-					101	Laptop				Electronics	1500	50			✓
ProductID	Name	Category	Price	Stock																																																												
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101	Laptop																																																															
Electronics	1500	50																																																														

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Create a new table named **products** with the following specifications:

- **product_id** as INTEGER and primary key.
- **product_name** as TEXT and not NULL.
- **list_price** as DECIMAL (10, 2) and not NULL.
- **discount** as DECIMAL (10, 2) with a default value of 0 and not NULL.
- A CHECK constraint at the table level to ensure:
 - **list_price** is greater than or equal to **discount**
 - **discount** is greater than or equal to 0
 - **list_price** is greater than or equal to 0

For example:

Test	Result
INSERT INTO products (product_id, product_name, list_price) VALUES (2, 'Product B', 50.00); SELECT * FROM products;	<pre> product_id product_name list_price discount ----- ----- -- 2 Product B 50 0 </pre>

Answer: (penalty regime: 0 %)

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```

CREATE TABLE products (
    product_id INTEGER PRIMARY KEY,
    product_name TEXT NOT NULL,
    list_price DECIMAL(10, 2) NOT NULL,
    discount DECIMAL(10, 2) NOT NULL DEFAULT 0,
    CHECK (
        list_price >= discount
        AND discount >= 0
        AND list_price >= 0
    )
);

```

	Test	Expected	Got	
✓	INSERT INTO products (product_id, product_name, list_price) VALUES (2, 'Product B', 50.00); SELECT * FROM products;	<pre> product_id product_name list_price discount ----- ----- -- 2 Product B 50 0 </pre>	<pre> product_id product_name list_price discount ----- ----- -- 2 Product B 50 0 </pre>	✓
✓	INSERT INTO products (product_id, product_name, list_price, discount) VALUES (3, 'Product C', 20.00, 30.00);	Error: CHECK constraint failed: products	Error: CHECK constraint failed: products	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Create a new table named **item** with the following specifications and constraints:

1. **item_id** as TEXT and as primary key.
2. **item_desc** as TEXT.
3. **rate** as INTEGER.
4. **icom_id** as TEXT with a length of 4.
5. **icom_id** is a foreign key referencing **com_id** in the **company** table.
6. The foreign key should set NULL on updates and deletes.
7. **item_desc** and **rate** should not accept NULL.

For example:

Test	Result
INSERT INTO item VALUES("ITM5","Charlie Gold",700,"COM4"); UPDATE company SET com_id='COM5' WHERE com_id='COM4'; SELECT * FROM item;	item_id item_desc rate icom_id ----- ----- ITM5 Charlie Gold 700

Answer: (penalty regime: 0 %)

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```
CREATE TABLE item (
    item_id TEXT PRIMARY KEY,
    item_desc TEXT NOT NULL,
    rate INTEGER NOT NULL,
    icom_id TEXT CHECK (LENGTH(icom_id) = 4),
    FOREIGN KEY (icom_id)
        REFERENCES company(com_id)
        ON UPDATE SET NULL
        ON DELETE SET NULL
);
```

	Test	Expected	Got	
✓	INSERT INTO item VALUES("ITM5","Charlie Gold",700,"COM4"); UPDATE company SET com_id='COM5' WHERE com_id='COM4'; SELECT * FROM item;	item_id item_desc rate icom_id ----- ----- ITM5 Charlie Gold 700	item_id item_desc rate icom_id ----- ----- ITM5 Charlie Gold 700	✓
✓	INSERT INTO item VALUES("ITM6","Pea-n-Nut",14,"COM2"); INSERT INTO item VALUES("ITM5","Charlie Gold",700,"COM4"); DELETE FROM company WHERE com_id='COM2'; SELECT * FROM item;	item_id item_desc rate icom_id ----- ----- ITM6 Pea-n-Nut 14 ITM5 Charlie Go 700 COM4	item_id item_desc rate icom_id ----- ----- ITM6 Pea-n-Nut 14 ITM5 Charlie Go 700 COM4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question 9

Correct

Mark 1.00 out of 1.00

Insert the below data into the **Employee** table, allowing the **Department** and **Salary** columns to take their default values.

EmployeeID	Name	Position
-----	-----	-----
4	Emily White	Analyst

Note: The Department and Salary columns will use their default values.

For example:

Test	Result		
SELECT EmployeeID, Name, Position FROM Employee;	EmployeeID	Name	Position
	-----	-----	-----
	4	Emily White	Analyst

Answer: (penalty regime: 0 %)

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```
INSERT INTO Employee (EmployeeID, Name, Position)
VALUES (4, 'Emily White', 'Analyst');
```

	Test	Expected	Got	
✓	SELECT EmployeeID, Name, Position FROM Employee;	EmployeeID Name Position ----- ----- -- ----- 4 Emily White Analyst	EmployeeID Name Position ----- ----- -- ----- 4 Emily White Analyst	✓
✓	SELECT count(*) FROM Employee ;	count(*) ----- 1	count(*) ----- 1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Question **10**

Correct

Mark 1.00 out of 1.00

Create a table named **Invoices** with the following constraints:

- **InvoiceID** as INTEGER should be the primary key.
- **InvoiceDate** as DATE.
- **Amount** as REAL should be greater than 0.
- **DueDate** as DATE should be greater than the **InvoiceDate**.
- **OrderID** as INTEGER should be a foreign key referencing **Orders(OrderID)**.

For example:

Test	Result
INSERT INTO Orders (OrderID, OrderDate, CustomerID) VALUES (1, '2024-08-01', 1);	InvoiceID InvoiceDate Amount DueDate OrderID
INSERT INTO Invoices (InvoiceID, InvoiceDate, Amount, DueDate, OrderID) VALUES (1, '2024-08-01', 100.0, '2024-09-01', 1);	----- ----- ----- -- -----
SELECT * FROM Invoices;	1 2024-08-01 100.0 2024-09-01 1

Answer: (penalty regime: 0 %)

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```
CREATE TABLE Invoices (  
    InvoiceID INTEGER PRIMARY KEY,  
    InvoiceDate DATE,  
    Amount REAL CHECK (Amount > 0),  
    DueDate DATE CHECK (DueDate > InvoiceDate),  
    OrderID INTEGER,  
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)  
);
```

	Test	Expected	Got	
✓	INSERT INTO Orders (OrderID, OrderDate, CustomerID) VALUES (1, '2024-08-01', 1); INSERT INTO Invoices (InvoiceID, InvoiceDate, Amount, DueDate, OrderID) VALUES (1, '2024-08-01', 100.0, '2024-09-01', 1); SELECT * FROM Invoices;	InvoiceID InvoiceDate Amount DueDate OrderID ----- - ----- --- 1 2024-08-01 100.0 2024-09-01 1	InvoiceID InvoiceDate Amount DueDate OrderID ----- - ----- --- 1 2024-08-01 100.0 2024-09-01 1	✓
✓	INSERT INTO Invoices (InvoiceID, InvoiceDate, Amount, DueDate, OrderID) VALUES (2, '2024-08-02', 20, '2024-09-02', 1);	Error: FOREIGN KEY constraint failed	Error: FOREIGN KEY constraint failed	✓

Passed all tests! ✓

